CPS610-W26, Assignment-2, Labs 3 – Database Systems-II **Due: Feb12, 2026, 4pm**

(The TA will evaluate your work and ask questions about progress and final results in the Lab sessions: on Feb11, Feb12)

Objectives

- Part1: Understanding and working with "PL/Sql" on Oracle Database Management System
- Part2: Understanding and working with user-defined type or "OBJECT" on Oracle Database System

Part1 Instructions: [Total 21 Marks]

In a Relational Database, using Procedural Language extensions to Sql or "PL/Sql" adds the procedural language constructs to the Sql language in order to cover its programming limitations. It will also improve the programming techniques in applications which run on databases using just Sql queries. It was designed to work only with Oracle DBMS and not working with any other DBs.

To start programming with PL/Sql, open a CLI on your system and login to your Oracle database (using system password set while installing Oracle). Then try the following code, which prints a welcome message on the display: (use a plain text editor to make and save the file as "X.sql"; run it at command prompt with "@ X.sql")

```
SQL> set serveroutput on
SQL> begin
  2  dbms_output.put_line('Welcome to Oracle PL-Sql');
  3  end;
  4  /
Welcome to Oracle PL-Sql

PL/SQL procedure successfully completed.

SQL>
```

Part2 Instructions:

In a Relational Database, "OBJECT" is a data structure to store or to refer to data. The most common object that is used in a database is "Table".
-An Object Type is the concept for defining an object, like a Table or Index.
-An Object Instance can be created based on an object type, like st_table that can be created as an instance of the object type Table.

To start programming with OBJECT, continue with the CLI you opened above on your system. Then try the following code, which creates an OBJECT, a user-defined type which is a "St_Rec", and then creates a Table named st_table based on the St_Rec. A message will be printed on display with each successful command:

```
SQL> Create TYPE St_Rec AS OBJECT
 2  (St_Name   VARCHAR(10),
3  St_Id      Number(7),
4  St_Email   VARCHAR(50));
5  /

Type created.

SQL> Create TABLE st_table of St_Rec;

Table created.

SQL>
```

Add the followings to the tables created in Assignment-1only by using PL/Sql:
[16 Marks], (2 marks for each step including screen-shots)

1- Create a new type named "depart_type" with this schema:
   Name, Faculty, Building, Phone
2- Create a new table named "Professor" with this schema:
   Name, Emp_id, Email, depart_type
   First make a type of this schema and then create the table based on that type.
   Note that deprt_type is itself a type.
3- Insert values in the "Professor" table at least for 5 records.
4- Display the whole "Professor" table on the screen.
5- Add a new attribute to the "Professor" table named "Income" and insert proper input values.
6- Write a for-loop to calculate and display the tax (30% of the Income) for all Professors.
7- Write a procedure to find and display all the professors' incomes which are less than $40,000.
8- Write a procedure to find and display the average of all the professors' incomes.

Note - Add screen-shots from each step with proper title in your submission.

9- Considering the whole system (definition of the relational tables in A1 plus the tables above), draw the class-diagram and its corresponding ER diagram. [5 Marks]

References:

https://www.oracle.com/ca-en/
https://w3resource.com/plsql-exercises/
https://www.oracletutorial.com/plsql-tutorial/
https://www.techonthenet.com/oracle/loops/if_then.php

Deliverables & Marking Schema

- Students are required to go through the instructions above, discuss their solution via team work, and finally submit their solution through D2L by the due date. Only ONE submission is required for each team.
- Teams are also required to indicate their partial work to the TA, asking their questions and getting directions from the TA during the lab times until the due.
- The TA will mark assignments by 1) individual assessment during the Lab time and 2) evaluation of the submission. Lab attendance is mandatory to get the both marks above for this assignment.
- The marks for Lab-3 (Assignment-2) will be based on the followings: **[Total 21 Marks = weight 3%]**
  1) Individual assessment in the Lab sessions on **Feb11** and **Feb12** (only for attended members in the Labs)
  2) Evaluation of the submission by **Feb12, 4pm** (only for attended members in the Labs)